

Sanjay Selvaraj

Site Reliability Engineer (SRE)

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Professional Summary

Site Reliability Engineer with hands-on experience in incident management, production reliability, Kubernetes troubleshooting, AWS cloud operations, observability, and Python automation. Managed 250+ operational incidents with a typical MTTR under 3 hours, led high-severity war room coordination with VP-level stakeholders, built internal automation tooling to reduce operational toil, and proactively drove reliability improvements through data-driven infrastructure changes. Experienced in on-call operations and cross-functional stakeholder coordination across distributed production environments.

Technical Skills

Cloud Platforms: AWS (EKS, EC2, IAM, CloudWatch, S3, RDS, Secrets Manager, Cross-Account Role Assumption)

Containers & Orchestration: Kubernetes, Docker, Argo CD, Kubernetes CronJobs, Helm (basic exposure), Service Accounts, Kubernetes Secrets, HPA (Horizontal Pod Autoscaler)

Monitoring & Observability: Splunk (Log Analysis), Dynatrace (APM & Metrics), BPM Synthetic Monitoring, Observability

Incident & Reliability Engineering: Incident Management, Problem Management, Root Cause Analysis (RCA), MTTR Optimization, On-call Operations, Toil Reduction, Production Support, Incident Coordination, War Room Coordination

CI/CD & Developer Tools: Jenkins, Git, Bitbucket, Jira, Linux, pytest, pre-commit hooks, uv package manager, linting, environment configuration

Programming & Engineering: Python, SQL, Java (code comprehension), JavaScript (basic), Microservices Troubleshooting, Distributed Systems Basics, Networking, OOP, DBMS, Operating Systems

Expanding expertise in: Bash scripting, Advanced AWS cloud operations

Professional Experience

ADP | Associate Software Engineer (SRE)

January 2025 – Present

Chennai | *Converted from Intern to Associate Software Engineer*

Incident Operations Management (IOM)

- Coordinated high-severity war room investigations involving 30–40+ participants across engineering, director, and VP-level stakeholders — driving resolution of critical production incidents.
- Maintained platform stability across 7+ production environments through proactive incident response, escalation management, and stakeholder coordination.
- Managed 250+ operational incidents maintaining a typical MTTR under 3 hours through structured alert ownership, rapid escalation, and coordinated resolution across engineering and operational teams.
- Sustained uninterrupted production coverage during alternate-week on-call rotations including weekends by serving as the primary responder for operational reliability across all active environments.
- Accelerated root cause identification by investigating operational issues using Splunk, Dynatrace, BPM synthetic monitoring, Kubernetes diagnostics, and AWS EKS troubleshooting.

Problem Management & Root Cause Analysis

- Analyzed 3 months of alert data to identify recurring OOM patterns across services — designed and merged HPA configurations to proactively address memory-related pod restarts before they escalate to incidents.
- Identified root causes for 10+ recurring incident patterns and collaborated with engineering teams to implement permanent fixes, improving overall platform reliability.
- Investigated JVM memory issues, thread dumps, pod restarts, readiness failures, timeout issues, and downstream dependency failures to improve service reliability.
- Improved service stability by proactively identifying insufficient memory allocation in scheduled workloads and driving permanent engineering fixes that eliminated repeat pod restart incidents.

Automation & Internal Tooling

Incident Management Bot

- Reduced manual incident coordination effort by ~10–15 minutes per incident by enhancing an internal automation platform integrating Jira, Webex, and Opsgenie workflow automation across engineering and operational teams.

AWS Health Event Notifier

- Improved visibility into AWS maintenance events and reduced avoidable on-call noise by building and production-deploying a Python tool that proactively alerts SRE, DevOps, and infrastructure stakeholders across infrastructure boundaries via cross-account IAM role assumption.
- Implemented deployment workflows using Jenkins CI, Argo CD, Kubernetes CronJobs, IAM role assumption, Kubernetes Secrets, and pytest to ensure production-grade quality.

RCA Investigation Assistant

- Reduced manual RCA investigation effort by automating log, metric, and incident-context correlation — replacing manual multi-source investigation with a single summarized output for faster engineering triage.

Achievements

- Converted from Intern to Associate Software Engineer at ADP based on demonstrated production impact.
- Published IEEE research paper: *Deep Learning with Multi-Modal Imaging: An Advanced Approach to Brain Tumor Detection*, IEEE, 2024.

Education

KPR Institute of Engineering and Technology, Coimbatore

Bachelor of Engineering – Computer Science and Engineering (2021–2025)

CGPA: 7.78/10